ALICE L WALTON SCHOOL OF MEDICINE

RBC Metabolism

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Review how hemoglobin affinity is altered by biological parameters (e.g. pH).
2. Explain the purpose of the metabolic reactions performed within mature erythrocytes.
3. Compare and contrast anemia caused by deficiencies in glucose 6-phosphate dehydrogenase and pyruvate kinase.
4. Outline the f[illegible] methemoglo[illegible]

CASE CONNECTION

A 60-year-old woman comes to the clinic complaining of worsening shortness of breath and blue skin discoloration. She recently started taking nitrates for angina and is worried she is having a heart attack.

Examination shows tachycardia and cyanosis. Pulse oximetry reads 85%, but arterial blood gas analyses show normal PaO_2. The condition is refractory to oxygen therapy. Additional measurements identify elevated methemoglobin levels.

What accounts for this patient's condition and which therapeutic is warranted?

GO TO CONCLUSION ↓

Review how hemoglobin affinity is altered by biological parameters (e.g. pH) (LO#1)

As a brief review, the predominate function of the red blood cell (RBC) entails the transport of oxygen from the lungs to the peripheral tissues. **Remember**, the affinity of Hb tetramers for oxygen can be altered to aid or hinder oxygen delivery depending upon bodily needs. This phenomenon is commonly plotted as an oxygen dissociation curve with p50 values representing relative affinity (Review Brick). Select examples of parameters that **decrease** hemoglobin (Hb) affinity for oxygen (*i.e. shift curve to the right causing a higher p50*) are summarized below for quick reference:

- **Low pH:** known as the Bohr effect in which a lower pH environment protonates select histidine residues to cause Hb to have a lower affinity for oxygen thereby, increasing O_2 delivery to tissues.
- **Elevated pCO_2:** CO_2 causes a shift in the bicarbonate blood buffer to produce more carbonic acid (H_2CO_3) to result in increased [H^+] producing the Bohr effect described above. To a smaller extent, enriching CO_2 also augments the formation of carbaminohemoglobin ($HbCO_2$) which has a decreased affinity for oxygen binding.
- **High temperatures:** Kinetic energy allows for augmented gas dissociation to aid a physically active muscle.
- **2,3-BPG:** 2,3-bisphosphoglycerate (2,3-BPG) is formed in RBCs from the glycolytic intermediate 1,3-bisphosphoglycerate by the enzyme bisphosphoglycerate mutase within the Rapoport-Luebering shunt to lower Hb affinity for oxygen. RBCs modulate Hb oxygen *rapidly* by favoring the synthesis or degradation of 2,3-BPG.

To fully understand RBC function as well as dysfunction, a discussion of erythrocyte metabolism is required. For example, notice how all of the bullet points above demonstrate that metabolism alone (e.g. low pH from lactate, high CO_2 formation from oxidative catabolism, 2,3-BPG from glycolysis) influences Hb affinity.

REVIEW QUES[illegible]BPG cause the oxygen dissociation curve of adult Hb (HbA) to shift?

CLINICAL REASONING

The molecules listed above are natural mechanisms that modulate Hb affinity for dynamic oxygen delivery to tissues. Remember, molecules such as carbon monoxide (i.e. CO) also alter Hb affinity but are considered poisons due to their unregulated, decremental change in hemoglobin affinity.

Explain the purpose of the metabolic reactions performed within mature erythrocytes (LO#2)

Mature erythrocytes lack all intracellular organelles including mitochondria, so the metabolic enzymes of the red blood cell are limited to those found in the cytoplasm. This impacts:

- **Energy production:** an RBC's ATP production is limited to substrate level phosphorylation via glycolysis. Lactate is produced from pyruvate to regenerate NAD^+ to provide glycolysis with the required coenzymes.
- **Oxidative stress**[illegible]lood cell contains enzymes necessary for the neutralization of reactive oxygen species.
- **Altering Hb affinity:** As described in the previous section, the glycolytic derivative 2,3-BPG is utilized to modulate Hb binding affinity. This molecule is only synthesized in red blood cells at appreciable levels specifically for this purpose.

CLINICAL REASONING

The reliance of RBCs on lactate production to sustain glycolysis is why lactate dehydrogenase (LDH) is enriched in RBCs and is used to confirm suspected hemolysis.

Critical Note: Due to the reliance on glycolysis, RBCs **must** intake blood glucose for proper function. Therefore, RBCs use GLUT1 to transport circulating glucose even at low concentrations that occur with prolonged fasting. While the source of the glucose might change (e.g. dietary, glycogen from liver, gluconeogenesis from liver), the identity of the fuel source (i.e. glucose) is not altered based upon dietary state. This unwavering use of circulating glucose is why RBCs lack ***both*** insulin and glucagon receptors. This metabolic aspect of RBCs is unique as most tissues adapt fuel utilization based upon glucose availability (**Table 1**).

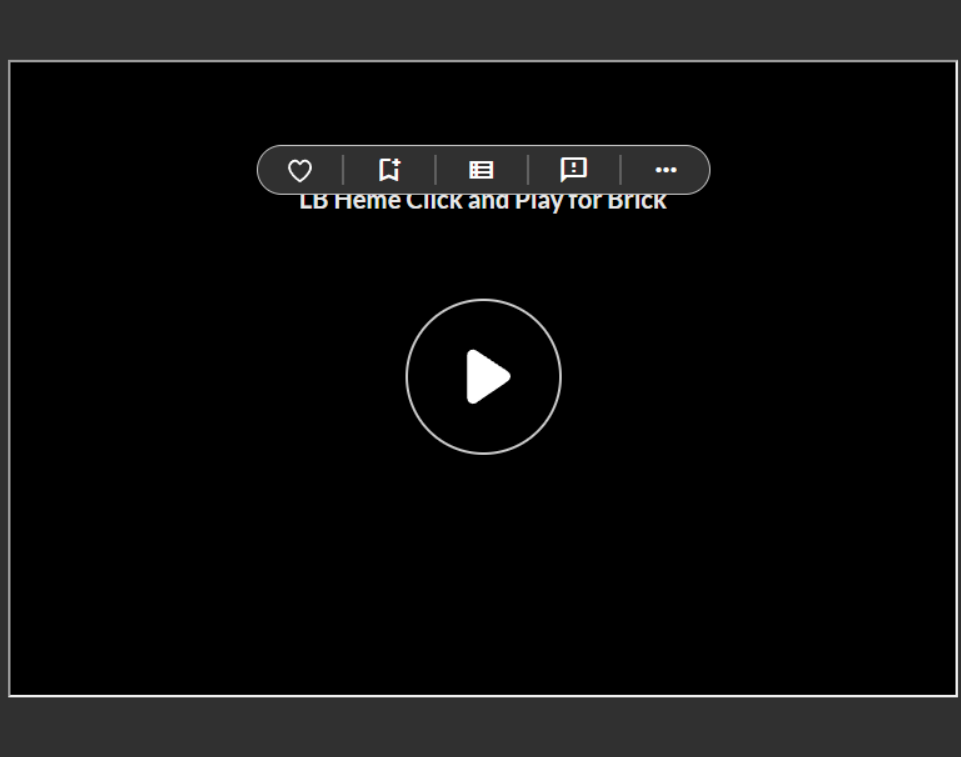
Organ	Well-fed	Fasting	Prolonged Fasting
Liver	predominately glucose	fatty acids	fatty acids
Skeletal Muscle (*resting*)	g[illegible]	[illegible]	more fatty acids, less ketone bodies
Cardiac muscle	fatty acids	fatty acids, ketone bodies	more fatty acids, less ketone bodies
Adipocytes	glucose	fatty acids	fatty acids
Brain	glucose	glucose	ketone bodies
RBCs	glucose	glucose	glucose

Table 1. Fuel utilization by tissue during distinct dietary states.

CLINICAL REASONING

To demonstrate the absolute importance of glycolysis within RBCs, consider the following. Despite only making up ~3% of the total body mass, RBCs contribute ~10% of total body glucose metabolism. This is because RBCs have the greatest rate of glucose utilization of any tissue (~ 4X greater). Interestingly, the impressive glycolytic rate observed with RBCs is coupled to one of the lowest bodily ATP synthesis rates. This is due to the fact that RBCs are acellular to allow the needed biconcave morphology. In the absence of mitochondria, the complete combustion of glucose is impossible, and instead, lactate is formed.

Perturbations in glycolysis routinely manifest in RBC dysfunction as RBC metabolism hinges on glucose oxidation. The video below briefly summarizes how known disruptions in this pathway manifest in select RBC disorders.



Compare and contrast anemia caused by deficiencies in glucose 6-phosphate dehydrogenase and pyruvate kinase (LO#3)

The deformable morphology of RBCs is ideally tailored for oxygen delivery (Review Brick) and conditions that compromise this structural feature cause disease. We will first consider the two distinct hemolytic anemias caused by aberrant RBC metabolism that cause irreversible RBC rupture or premature cellular clearance. Commonly shared features of hemolysis irrespective of underlying cause can include:

- hyperbilirubinemia, jaundice and darkened urine
- decreased serum haptoglobin
- increased serum LDH
- fatigue
- increased reticulocyte count
- varying severity dependent upon variant
- varying age of onset (neonatal period through adulthood)

Glucose 6-phosphate dehydrogenase (G6PD) deficiency

In BOOM, we considered the pentose phosphate pathway (PPP), a glycolytic shunt that yields NADPH and ribose 5-phosphate. NADPH is generated in the first reaction of the PPP catalyzed by glucose 6-phosphate dehydrogenase (G6PD). Many cells such as adipocytes and hepatocytes also generate NADPH through alternative pathways apart from PPP (i.e. malic enzyme). However, the PPP is the **sole source of NADPH** in mature RBCs making erythrocytes particularly vulnerable to decreased NAPDH. In the RBC, the NADPH provides reduction power critical for eliminating harmful oxidative species such as hydrogen peroxide or hydroxyl radicals generated during oxygen exchange (**Figure 1**).

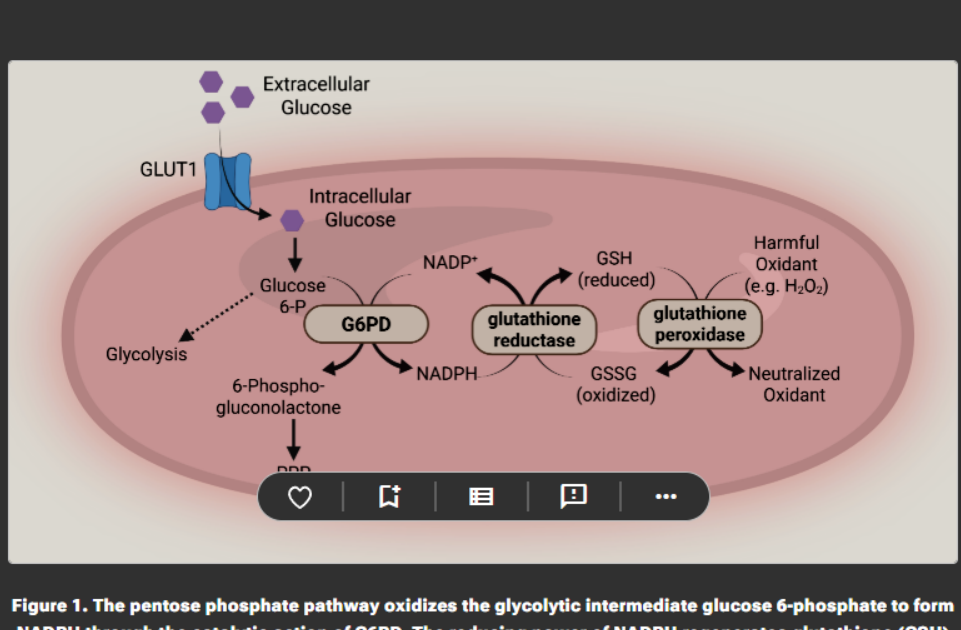


Figure 1. The pentose phosphate pathway oxidizes the glycolytic intermediate glucose 6-phosphate to form NADPH through the catalytic action of G6PD. The reducing power of NADPH regenerates glutathione (GSH) from oxidized glutathione dimers (GSSG) through glutathione reductase. Reduced glutathione (GSH) neutralizes damaging oxidants via glutathione peroxidase. If this step is prevented, hemolysis occurs due to lipid and protein damage caused by reactive oxygen species (ROS).

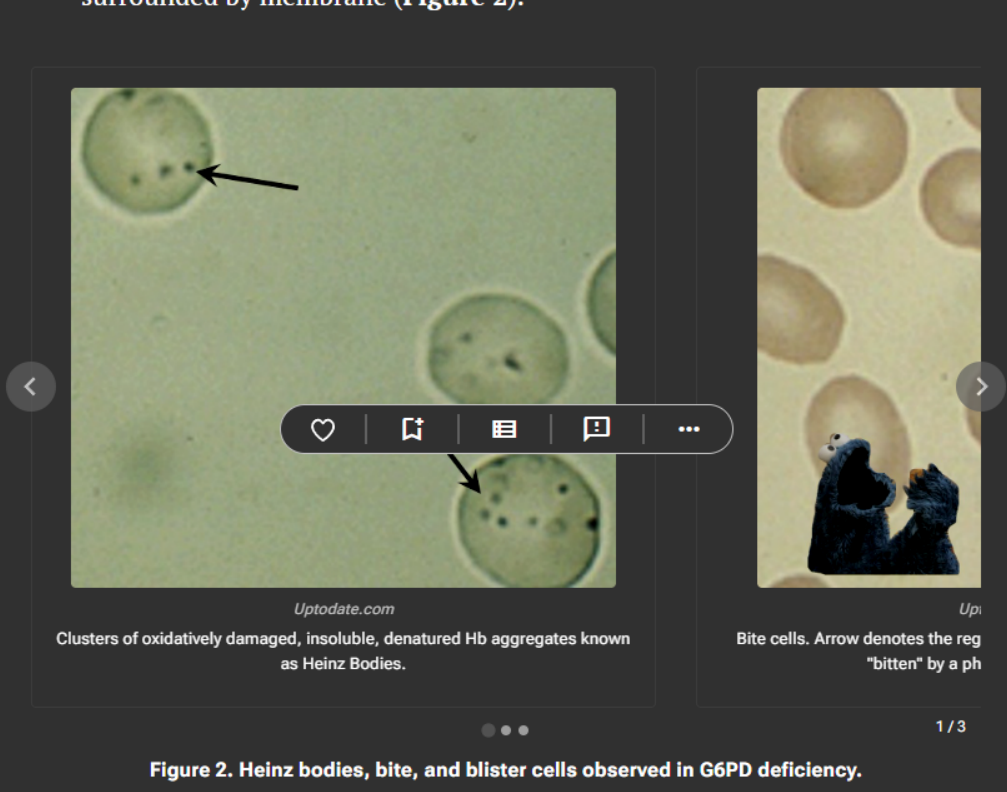
G6PD deficiency is the most common enzymopathy within humans, and thereby default, it is also the most prevalent inherited RBC enzyme deficiency that causes hemolytic anemia. The underlying basis of the cellular damage is **oxidative stress**. In many cases, patients with G6PD variants are asymptomatic until exposed to a trigger. Below are a few of the common factors that can initiate or potentiate hemolysis due to oxidative stress:

- **infections**: most common hemolytic trigger in G6PD deficiency
- **select drugs**: broad categories that include select antipyretics, antimalarial, and antibiotics
- **fava beans**: not all individuals with G6PD deficiency will exhibit favism but all persons with favism have G6PD deficiency

Critical Note: Catalase, much like glutathione peroxidase, converts hydrogen peroxide into water as a protective measure against oxidative stress. While catalase is highly enriched in the liver, RBCs also express this detoxification enzyme. However, impairment of the glutathione antioxidant defense mechanisms due to a lack of NADPH generation is sufficient to cause hemolytic disease. This is partly due to the fact that glutathione peroxidase, unlike catalase, is utilized to neutralize a spectrum of oxidative species such as lipid peroxides and no[illegible]ulation of multiple reactive oxygen specie[illegible]ated with G6PD deficiency.

Features that can distinguish G6PD deficiency from other anemias are:

- **Heinz bodies**: these are oxidatively damaged, insoluble aggregates of Hb protein (**Figure 2**). Heinz bodies tend to localize around the plasma membrane causing it to become less flexible and reducing its deformability. This structural change promotes lysis due to the increased mechanical stress and strain within the vasculature.
- **Bite cells:** as Heinz bodies form, they are perceived as non-self or foreign. This prompts an immune response from splenic, phagocytic macrophages that selectively remove "chunks" of the membrane and cytoplasm that resembles a bite out of a cookie (**Figure 2**).
- **Blister cells:** The same oxidative damage to Hb that prompts other features described above also cause blister cells. Damaged Hb can partition/cluster to one side of the cell causing a retraction of color on the opposite side. The regional clearing appears like a pale, fluid-filled blister surrounded by membrane (**Figure 2**).


Uptodate.com
Clusters of oxidatively damaged, insoluble, denatured Hb aggregates known as Heinz Bodies.

Up
Bite cells. Arrow denotes the reg
"bitten" by a ph

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Figure 2. Heinz bodies, bite, and blister cells observed in G6PD deficiency.

What comprises Heinz bodies?

CLINICAL REASONING

There are over 400 distinct point mutations noted in the literature for G6PD deficiency. There ar[illegible]inding affinity for $NADP^+$ or gross structural instability that account for disrupted G6PD function.

In many cases, newly released reticulocytes contain **unstable but kinetically normal G6PD** as the enzyme has been recently synthesized. Older RBCs typically have much lower G6PD activity and are more prone to oxidative stress and lysis. Remember, mature RBCs lack a nucleus so replacing defective G6PD with freshly translated, functional protein is not an option as RBCs age.

The burst in reticulocyte formation coupled with the turnover of older, defective RBCs causes the hemolytic episodes to be self-limiting. This dynamic informs G6PD testing guidelines. Profiling G6PD activity around a hemolytic crisis when reticulocyte count is most likely high can generate falsely elevated G6PD activity not indicative of G6PD deficiency. This is why testing guidelines include a 3-month (i.e. 120 day) testing window from a hemolytic event as this span of time is the average lifespan of an RBC.

Pyruvate kinase deficiency

Other metabolic issues apart from G6PD deficiency within erythrocytes cause hemolytic anemia. An inherited deficiency in erythrocyte **pyruvate kinase** leads to hemolytic anemia, and it is the second most common RBC-specific enzyme disorder. As RBCs are fully reliant upon glycolysis for ATP production, deficiency in pyruvate kinase is devastating to normal RBC function as this step accounts for 50% of the ATP yield from glycolysis.

The exact mechanism of how ATP deprivation culminates in RBC lysis is still largely theoretical. However, ATP is definitively required for:

- maintaining proper ion balance across the cell membrane (Na^+, K^+, and Ca^{++})
- phosphorylating membrane proteins
- priming reactions of glycolysis (i.e. steps 1 and 3)

In the absence of sufficient ATP, the current model posits that the RBC becomes inflexible and this deviation from normal morphology accounts for turnover and lysis (**Figure 3**). Briefly, RBC ion transporters such as the K^+/Na^+ pump are unable to function effectively due to the lack of ATP. An inability to maintain ion gradient promotes cellular water loss which increases the intracellular Hb concentration causing the cell to abnormally shrink. RBC membrane retraction forms erratic membrane protrusions. These changes effectively increase cytosolic viscosity which likely reduce RBC membrane flexibility making the affected RBCs more susceptible to damage by shear forces present in the circulation. The damaged RBCs are selectively removed from circulation, leading to the anemia.

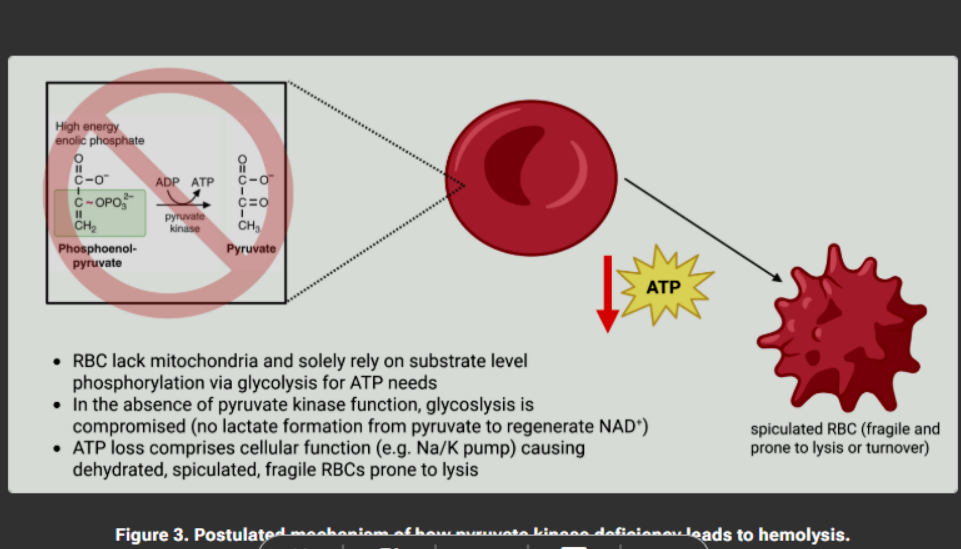

High energy enolic phosphate
Phosphoenol-pyruvate
ADP ATP
pyruvate kinase
Pyruvate
ATP
RBC lack mitochondria and solely rely on substrate level phosphorylation via glycolysis for ATP needs
In the absence of pyruvate kinase function, glycolysis is compromised (no lactate formation from pyruvate to regenerate NAD+)
ATP loss comprises cellular function (e.g. Na/K pump) causing dehydrated, spiculated, fragile RBCs prone to lysis
spiculated RBC (fragile and prone to lysis or turnover)

Figure 3. Postulated mechanism of how pyruvate kinase deficiency leads to hemolysis.

An interesting aspect of pyruvate kinase deficiency is that it is associated with **~3-fold elevation in 2,3-BPG** concentration that results from the blockage of the conversion of phosphoenolpyruvate to pyruvate. While not a clinical diagnostic, it is postulated elevated 2,3-BPG levels aids the condition by increasing oxygen delivery. Because 2,3-BPG binding to hemoglobin decreases its affinity for oxygen, the RBCs that remain in circulation are highly efficient in releasing their bound oxygen to the tissues. This does not rectify the condition but is likely an adaptive mechanism to alleviate severity.

Critical Note: Pyruvate kinase deficiency presents with **chronic hemolysis** at an early age with average diagnosis occurring before 6 months of age with either enzymatic activity assay or genetic testing. This is in contrast to the most common G6PD variants that cause **episodic hemolysis** that typically require an oxidative stress trigger to manifest.

CLINICAL CORRELATION

The clinical presentation of patients with PK deficiency is highly variable, ranging from a mild, fully compensated chronic hemolysis without anemia to life-threatening transfusion-dependent hemolysis present at birth. Patients with severe hemolysis may develop complications of chronic hemolytic stages, such as extramedullary hematopoiesis, pigmented gallstones comprised of bilirubin, and iron overload.

Management can include splenectomy due to increased risk of splenomegaly. In many instances, a cholecystectomy will be performed at the same time due to heightened likelihoo[illegible]

Pyruvate kinase forms ATP and which final product of glycolysis?

Outline the formation of methemoglobin and how methemoglobinemia is normally prevented (LO#4)

Methemoglobin (metHb) forms when the iron within the porphyrin ring is oxidized from the ferrous [Fe^{2+}] to the ferric [Fe^{3+}] form. MetHb forms spontaneously within RBCs during oxygen exchange at a low rate in all individuals. A normal methemoglobin fraction is about ~0-3% measured by co-oximetry.

Congenital and acquired conditions can lead to elevated levels of MetHb that warrant intervention. As metHb with ferric iron is **unable to bind and transport oxygen** pr[illegible]en delivery leading to tissue hypoxia and cyanosis. With mixed forms, a single Hb usually contains a mix of ferric and ferrous iron states across the tetramer which greatly impacts oxygen delivery. Mechanistically, the Hb with only ferric iron centers do not bind oxygen. However, when mixed, the ferric forms alter tetrameric Hb conformation resulting in increased oxygen affinity of the remaining, normal ferrous heme.

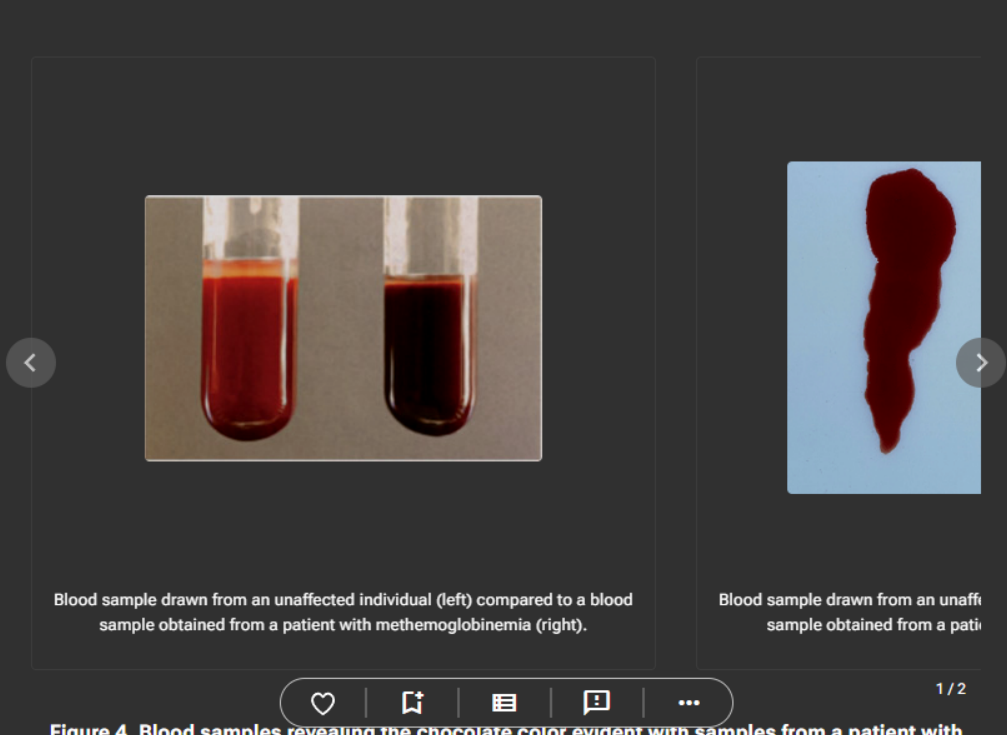
Symptom severity is directly proportional to the level of methemoglobin and the resulting extent of tissue hypoxia. While not everyone has symptoms, nearly all people with this condition have skin, nails or lips that are a distinctive shade of blue or purple. A hallmark is **"chocolate brown"** blood due to the enriched deoxygenated state that has been noted with metHb levels as low as 15% (**Figure 4**). Symptoms associated with higher levels of methemoglobin are generally as follows:

% MetHb	Symptoms
<15	Typically asymptomatic
15-30	Cyanosis, anxiety, light-headedness, fatigue, headache
30-50	Tachypnea, confusion, syncope
50-70	Seizures, arrhythmias, metabolic acidosis, coma
>70	Death

Table 2. Symptoms associated with varying metHb levels.

CLINICAL CORRELATION

A defining character[illegible] cyanosis", a brownish-grey-blue color of the skin and blood due to the heightened deoxygenated state of Hb within RBCs. The cyanosis caused by methemoglobinemia is **unresponsive to O_2** (i.e. no improvement in oxygen saturation with oxygen therapy). Importantly, paO_2 will be normal but O_2 saturation will typically be low with methemoglobinemia. However, pulse oximetry measurements **are not reliable** to determine the severity as methemoglobin interferes with the wavelength-based detection. A typical pulse oximetry value for methemoglobinemia will be ~85% irrespective of actual saturation due to the metHb interference. Co-oximetry is required to obtain accurate methemoglobin levels.


Blood sample drawn from an unaffected individual (left) compared to a blood sample obtained from a patient with methemoglobinemia (right).

Blood sample drawn from an unaffi
sample obtained from a pati

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Figure 4. Blood samples revealing the chocolate color evident with samples from a patient with methemoglobinemia.

Methemoglobin levels are usually maintained below 1% through the catalytic action of cytochrome-b5 reductase (**Figure 5**). Cytochrome-b5 reductase utilizes NADH formed in step 6 of glycolysis by glyceraldehyde 3-phosphate dehydrogenase (GAPDH) to reduce metHb back to functional hemoglobin.

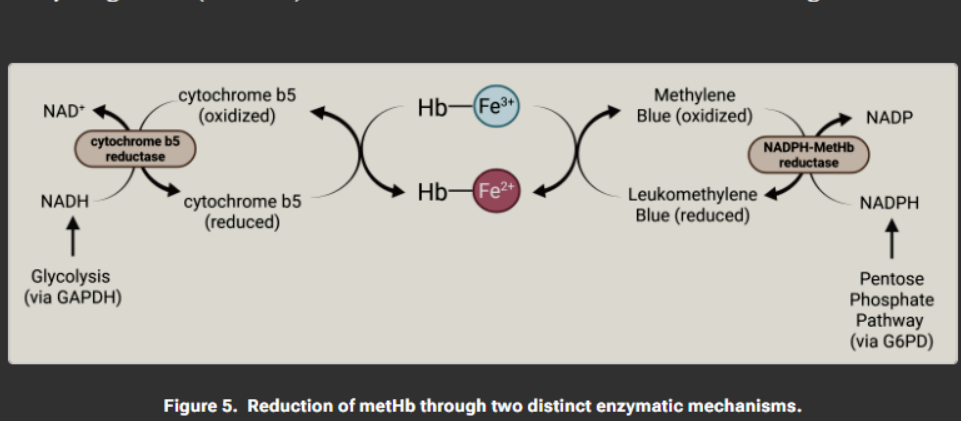

Figure 5. Reduction of metHb through two distinct enzymatic mechanisms.

Acquired vs Congenital forms of Methemoglobinemia

Acquired forms of methemoglobinemia usually arise due to exposure to chemical oxidants including nitrates/nitrites, quinones, aniline, and sulfonamides. Many such compounds are administered as therapeutics (e.g. anesthetics like benzocaine or lidocaine and dapsone, an anti-infective skin agent). Exposure can also occur from water and other environmental sources (e.g. nitrate-based fertilizers). Infants are particularly susceptible to this adverse condition as [illegible]tochrome b5 reductase as adults, and the iron within HbF is structurally more susceptible to oxidation than HbA.

CLINICAL CORRELATION

Benzocaine is applied topically through many over the counter sprays for sore throats. Overuse of throat spray can result in methemoglobinemia, and patients can become cyanotic. It is contraindicated in patients with decreased lung function or children under the age of 2 for these concerns.

Acquired forms are treated with antioxidants such as vitamin C (i.e. ascorbic acid) or methylene blue. Methylene blue is the typical treatment for acquired methemoglobinemia. It is a dark pigment historically used in textile fabric staining but was found to serve as an electron carrier in biochemical reactions. The NADPH-methemoglobin reductase enzyme provides an alternative pathways for reduction of metHb apart from cytochrome b5 reductase (**Figure 5**). While naturally expressed in RBCs, this enzyme does not contribute appreciably to metHb reduction under normal conditions. Fortunately, this pathway can be exogenously stimulated during times of high oxidative stress through therapeutic administration of methylene blue. Importantly, as the enzyme name indicates, NADPH-MetHb reductase utilizes the reducing power of NADPH formed by glucose-6-phosphate dehydrogenase (G6PD) within the PPP.

CLINICAL REASONING

A common misconception is that the G6PD deficiency which causes less NADPH yield contributes to increased risk for methemoglobinemia. But as stated above, the NADPH-MetHb reductase contributes minimally to the prevention of metHb accumulation as compared to cytochrome b5 reductase. Yet, the oxidative stressors that prompt hemolytic episodes in G6PD deficiency can indeed increase metHb levels.

A critical **contraindication** of methylene blue occurs for G6PD deficient patients. Treatment of methemoglobinemia with methylene blue is relatively discouraged in such scenarios because the reduction of methemoglobin is NADPH-dependent with methylene blue. As these patient's already have a diminished NADPH pool, stimulating an additional pathway requiring NADPH within the RBC is thought to potentiate hemolysis.

Congenital forms of methemoglobinemia are due to defects in the enzyme cytochrome b5 reductase or mutations in the genes that encode for globin proteins collectively known as hemoglobin M (HbM). In HbM, various mutations compromise the structural integrity of the heme binding pocket. These structural changes allow for greater auto-oxidation of the iron to ultimately increase metHb abundance. In contrast to acquired forms, methylene blue is typically not as effective for congenital forms and overall management is supportive in nature except in advanced cases.

CLINICAL CORRELATION

While typically associ[illegible]n is a therapeutic strategy to counteract the deleterious effects of cyanide (CN) poisoning. Amyl nitrite or sodium nitrite are administered to prompt oxidative changes to favor metHb formation. The ferric iron within metHb covalently coordinates the poisonous CN to promote excretion by the generation of cyanomethemoglobin. Binding to metHb prevents binding to complex IV of the ETC to provide intervention and restoration of ATP production.

Clearly, the side effects of this approach entail methemoglobinemia (e.g. headache, dizziness, etc.). If sufficient metHb is formed, it compromises oxygen delivery which may exacerbate, instead of rectify, metabolic disruptions from CN poisoning. The myriad of potential and concerning issues is why hydroxocobalamin is the first line antidote instead of nitrite administration.

ACTIVE LEARNING QUESTION

Match the enzyme with the correct descriptor

bisphosphoglycerate mutase	✓ forms a glucose derivative that decreases Hb affinity for oxygen
cytochrome b5 reductase	✓ uses reductive power formed in glycolysis to prevent metHb accumulation
glucose 6-phosphate dehydrogenase	✓ deficiency increases oxidative stress causing hemolysis
pyruvate kinase	✓ deficiency causes energy imbalances causing hemolysis
NADPH-MetHb reductase	✓ activated with methylene blue to prevent metHb accumulation

Explanation

See sections above.

CASE CONNECTION

BACK TO INTRODUCTION ↑

As the patient has impaired oxygen delivery due to oxidation of the iron within the heme ring, administration of methylene blue is recommended. This activates an alternative NADPH-dependent pathway to reduce the nonfunctional metHb back to Hb, restoring oxygen delivery.

Summary

- Hemoglobin affinity for oxygen is decreased by factors such as low pH, high pCO_2, elevated temperature, and increased 2,3-BPG.
- Mature erythroc[illegible]lytic shunts for ATP production and oxidative balance, respectively.
- Glucose 6-phosphate dehydrogenase deficiency causes hemolytic anemia due to oxidative stress, characterized by Heinz bodies, bite cells, and blister cells.
- Pyruvate kinase deficiency leads to hemolytic anemia through ATP deprivation, resulting in inflexible RBCs.
- Methemoglobin forms when iron in hemoglobin is oxidized from +2 to +3 which impairs oxygen delivery. This is naturally rectified by cytochrome-b5 reductase.
- Congenital methemoglobinemia results from enzyme defects, while acquired forms arise from chemical exposure, with methylene blue used for treatment except in G6PD deficiency.

Review Questions

1. Which metabolic pathway is absent within mature erythrocytes?

A. ○ glycolysis
B. ○ pentose phosphate pathway
C. ● beta-oxidation
D. ○ lactate formation

Hide Explanation

(C) is correct as mature erythrocytes lack mitochondria. (A), (B), and (D) all occur within mature RBCs.

2. Which deficiency increases methemoglobinemia risk?

A. ○ G6PD deficiency
B. ● cytochrome b5 reductase deficiency
C. ○ pyruvate kinase deficiency
D. ○ bisphosphoglycerate mutase deficiency

Hide Explanation

(B) is correct as congenital cytochrome b5 reductase deficiency prevents reduction of metHb. (A) is incorrect as it adversely impacts oxidative balance in distinct ways without increasing methemoglobinemia risk. (C) is incorrect as it impacts ATP production not oxidative stress. (D) is incorrect as this enzyme produced 2,3-BPG to alter Hb affinity naturally.

3. Which metabolic reaction decreases Hb affinity for oxygen?

A. ● generation of lactate
B. ○ G6PD defi[illegible]
C. ○ decreased 2,3-BPG levels
D. ○ hyperventilation

Hide Explanation

(A) is correct as lactate production produces protons to decrease pH leading to the Bohr effect. (B) is incorrect as this impacts oxidative stress but it does not alter Hb's affinity for oxygen. (C) and (D) are incorrect as they both lead to increased Hb affinity for oxygen.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Review how hemoglobin affinity is altered by biological parameters (e.g. pH).~~
02 ☑ ~~Explain the purpose of the metabolic reactions performed within mature erythrocytes.~~
03 ☑ ~~Compare and contrast anemia caused by deficiencies in glucose 6-~~

Bohr effect. (B) is incorrect as this impacts oxidative stress but it does not alter Hb's affinity for oxygen. (C) and (D) are incorrect as they both lead to increased Hb affinity for oxygen.

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03 ☑ ~~Compare and contrast anemia caused by deficiencies in glucose 6-phosphate dehydrogenase and pyruvate kinase.~~
04 ☑ ~~Outline the formation of methemoglobin and how methemoglobinemia is normally prevented.~~

Objective Confidence: 4/4